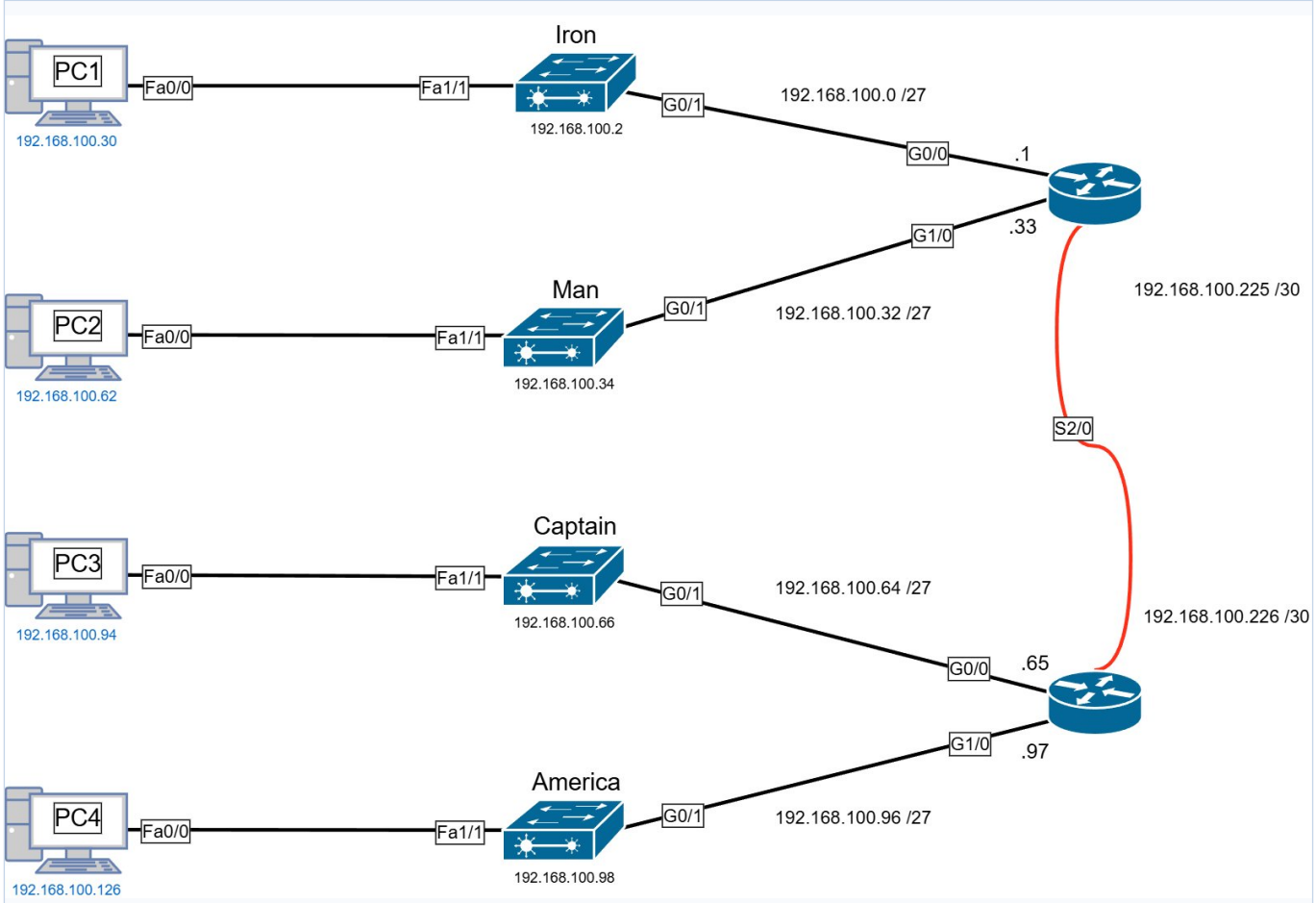
Descricao do Lab: Rede com dois roteadores (MAR e VEL), quatro switches de acesso (Iron, Man, Captain, America) e quatro hosts (PC1–PC4). Sub-redes /27 para LANs e /30 para link WAN serial. Roteamento estatico, verificacao de interfaces e testes de conectividade completos entre todas as sub-redes.

Dispositivo	Funcao	Enderecos Principais
MAR	Roteador principal	Gi0/0: 192.168.100.1 | Gi1/0: 192.168.100.33 | Se2/0: 192.168.100.225
VEL	Roteador secundario	Gi0/0: 192.168.100.65 | Gi1/0: 192.168.100.97 | Se2/0: 192.168.100.226
PC1/PC2	Hosts VLAN .0 e .32	PC1: .30 | PC2: .62 (sub-redes /27)
PC3/PC4	Hosts VLAN .64 e .96	PC3: .94 | PC4: .126 (sub-redes /27)
Link WAN	Serial ponto-a-ponto	192.168.100.224/30 — MAR:.225 <-> VEL:.226

1. Topologia da Rede



Topologia completa — MAR, VEL, Iron, Man, Captain, America | Link WAN Serial /30 (draw.io)

2. Calculo de Sub-Redes (/27 e /30)

Mascara /27 — 30 hosts utilizaveis por sub-rede
`Analise binaria: 255.255.255.224`
✔ *2^5 - 2 = 30 hosts | 8 sub-redes possiveis a partir de /24*

APROVADO

VLSM — Link WAN com /30 (economia de enderecos)
`Interface Serial2/0`
✔ *2^2 - 2 = 2 hosts — ideal para link ponto-a-ponto*

APROVADO

Sub-rede	Endereco de Rede	Mascara	1o Host	Ultimo Host	Broadcast	Uso
1	192.168.100.0	255.255.255.224	192.168.100.1	192.168.100.30	192.168.100.31	LAN Iron/MAR
2	192.168.100.32	255.255.255.224	192.168.100.33	192.168.100.62	192.168.100.63	LAN Man/MAR
3	192.168.100.64	255.255.255.224	192.168.100.65	192.168.100.94	192.168.100.95	LAN Captain/VEL
4	192.168.100.96	255.255.255.224	192.168.100.97	192.168.100.126	192.168.100.127	LAN America/VEL
WAN	192.168.100.224	255.255.255.252	192.168.100.225	192.168.100.226	192.168.100.227	Link MAR-VEL

```
Representacao binaria da mascara /27:
11111111 . 11111111 . 11111111 . 11100000
  255    .   255    .   255    .   224
Bits de rede: 27 (fixos) | Bits de host: 5 (variaveis)
Incremento entre sub-redes: 2^5 = 32
```

2.1. Planilha de Calculo — Todas as Sub-Redes /27 do bloco /24

A tabela abaixo foi elaborada manualmente para mapear **todas as 8 sub-redes /27 possiveis** dentro do bloco 192.168.100.0/24. O laboratorio utiliza as 4 primeiras sub-redes (redes .0, .32, .64 e .96) para os segmentos LAN, e a sub-rede .224 para o link WAN /30. As demais ficam disponiveis para expansao futura.

Cálculo de Sub-Redes para 25 hosts

Rede	Host	Broadcast
192.168.100.0	192.168.100.1 - 192.168.100.30	192.168.100.31
192.168.100.32	192.168.100.33 - 192.168.100.62	192.168.100.63
192.168.100.64	192.168.100.65 - 192.168.100.94	192.168.100.95
192.168.100.96	192.168.100.97 - 192.168.100.126	192.168.100.127
192.168.100.128	192.168.100.129 - 192.168.100.158	192.168.100.159
192.168.100.160	192.168.100.161 - 192.168.100.190	192.168.100.191
192.168.100.192	192.168.100.193 - 192.168.100.222	192.168.100.223
192.168.100.224	192.168.100.225 - 192.168.100.254	192.168.100.255

Calculo de Sub-Redes para 25 hosts — todas as 8 sub-redes /27 do bloco 192.168.100.0/24 (Notion)

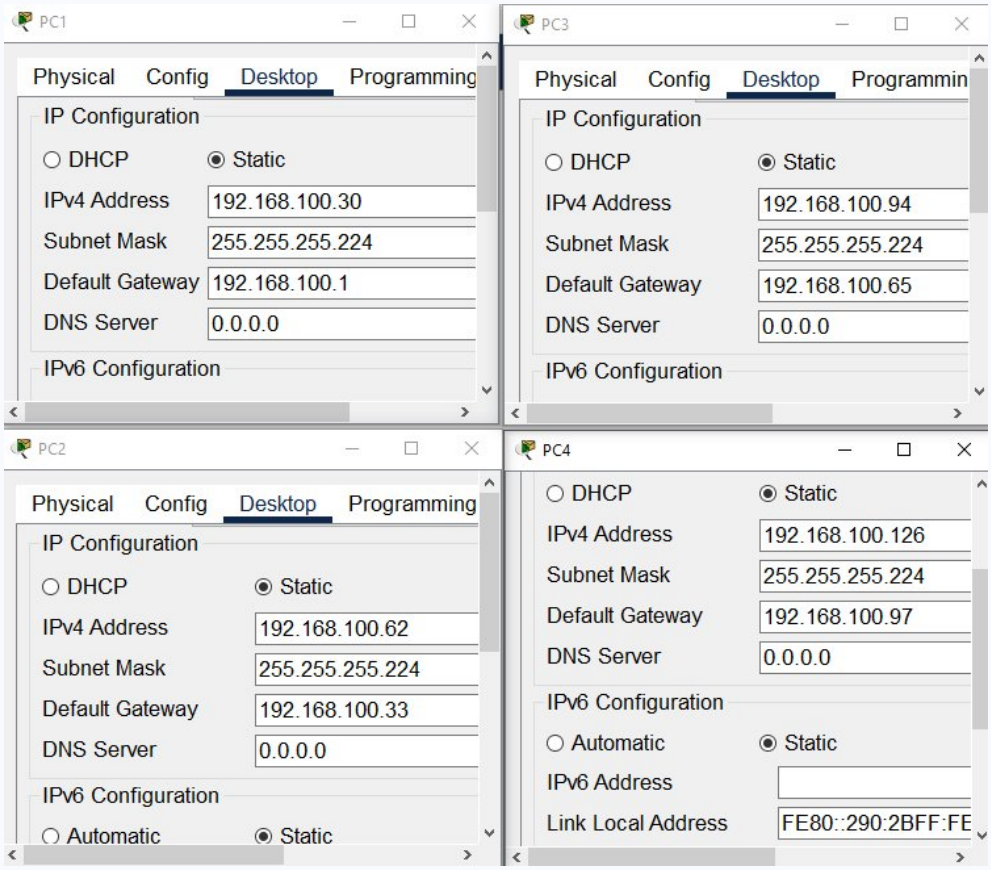
3. Configuracao dos PCs

PC1 — IP estatico na sub-rede 192.168.100.0/27
`Desktop > IP Configuration > Static`
✔ *IPv4: 192.168.100.30 | Mask: 255.255.255.224 | GW: 192.168.100.1*
APROVADO

PC2 — IP estatico na sub-rede 192.168.100.32/27
`Desktop > IP Configuration > Static`
✔ *IPv4: 192.168.100.62 | Mask: 255.255.255.224 | GW: 192.168.100.33*
APROVADO

PC3 — IP estatico na sub-rede 192.168.100.64/27
`Desktop > IP Configuration > Static`
✔ *IPv4: 192.168.100.94 | Mask: 255.255.255.224 | GW: 192.168.100.65*
APROVADO

PC4 — IP estatico na sub-rede 192.168.100.96/27
`Desktop > IP Configuration > Static`
✔ *IPv4: 192.168.100.126 | Mask: 255.255.255.224 | GW: 192.168.100.97*
APROVADO



IP Configuration dos quatro PCs — enderecos estaticos, mascara /27 e gateways corretos

4. Configuracao dos Roteadores (running-config)

MAR — Gi0/0 (.1) e Gi1/0 (.33) — sub-redes .0 e .32

`MAR# show running-config | section interface`

✔ *Interfaces GigabitEthernet configuradas com /27 — rotas estaticas para .64 e .96*

APROVADO

```
interface GigabitEthernet0/0
 ip address 192.168.100.1 255.255.255.224
 duplex auto
 speed auto
!
interface GigabitEthernet1/0
 ip address 192.168.100.33 255.255.255.224
 duplex auto
 speed auto
!
interface Serial2/0
 ip address 192.168.100.225 255.255.255.252
 clock rate 2000000
!
ip classless
ip route 192.168.100.64 255.255.255.224 192.168.100.226
ip route 192.168.100.96 255.255.255.224 192.168.100.226
```

Running-config do Roteador MAR — interfaces Gi0/0, Gi1/0, Se2/0 e rotas estaticas

VEL — Gi0/0 (.65) e Gi1/0 (.97) — sub-redes .64 e .96

APROVADO

`VEL# show running-config | section interface`

✔ *Interfaces GigabitEthernet configuradas com /27 — rotas estaticas para .0 e .32*

```
interface GigabitEthernet0/0
 ip address 192.168.100.65 255.255.255.224
 duplex auto
 speed auto
!
interface GigabitEthernet1/0
 ip address 192.168.100.97 255.255.255.224
 duplex auto
 speed auto
!
interface Serial2/0
 ip address 192.168.100.226 255.255.255.252
!
router rip
!
ip classless
ip route 192.168.100.0 255.255.255.224 192.168.100.225
ip route 192.168.100.32 255.255.255.224 192.168.100.225
```

Running-config do Roteador VEL — interfaces Gi0/0, Gi1/0, Se2/0 e rotas estaticas

5. Rotas Estaticas (show ip route)

MAR — Gateway de last resort nao definido (rede interna)

APROVADO

`MAR# show ip route`

✔ *C: .0/27 e .32/27 conectadas diretamente | S: .64 e .96 via 192.168.100.226*

```
     192.168.100.0/24 is variably subnetted, 5 subnets, 2
masks
C       192.168.100.0/27 is directly connected,
GigabitEthernet0/0
C       192.168.100.32/27 is directly connected,
GigabitEthernet1/0
S       192.168.100.64/27 [1/0] via 192.168.100.226
S       192.168.100.96/27 [1/0] via 192.168.100.226
C       192.168.100.224/30 is directly connected, Serial2/0
```

show ip route no MAR — C=diretamente conectada | S=rota estatica | 5 subnets, 2 masks

VEL — Rotas para sub-redes do MAR via 192.168.100.225

APROVADO

`VEL# show ip route`

✔ *C: .64/27 e .96/27 conectadas diretamente | S: .0 e .32 via 192.168.100.225*

```
     192.168.100.0/24 is variably subnetted, 5 subnets, 2 masks
S       192.168.100.0/27 [1/0] via 192.168.100.225
S       192.168.100.32/27 [1/0] via 192.168.100.225
C       192.168.100.64/27 is directly connected,
GigabitEthernet0/0
C       192.168.100.96/27 is directly connected,
GigabitEthernet1/0
C       192.168.100.224/30 is directly connected, Serial2/0
```

show ip route no VEL — C=diretamente conectada | S=rota estatica | 5 subnets, 2 masks

6. Interfaces dos Roteadores (show ip interface brief)

MAR — Gi0/0, Gi1/0 e Se2/0 — todas up/up

`MAR# show ip interface brief`

✔ *GigabitEthernet0/0: 192.168.100.1 — up/up | Gi1/0: .33 — up/up | Se2/0: .225 — up/up*

APROVADO

```
MAR#show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0     192.168.100.1   YES manual up
up
GigabitEthernet1/0     192.168.100.33  YES manual up
up
Serial2/0              192.168.100.225 YES manual up
up
```

show ip interface brief — MAR — todas as interfaces em estado up/up

VEL — Gi0/0, Gi1/0 e Se2/0 — todas up/up

`VEL# show ip interface brief`

✔ *GigabitEthernet0/0: 192.168.100.65 — up/up | Gi1/0: .97 — up/up | Se2/0: .226 — up/up*

APROVADO

```
VEL#show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0     192.168.100.65  YES manual up
up
GigabitEthernet1/0     192.168.100.97  YES manual up
up
Serial2/0              192.168.100.226 YES manual up
up
```

show ip interface brief — VEL — todas as interfaces em estado up/up

7. Testes de Conectividade (ping)

Ping inter-sub-rede — PC1 -> PC2 (192.168.100.62) — 3/4 — OK

`C:\> ping 192.168.100.62`

✔ *Reply from 192.168.100.62 — TTL=127 — roteamento MAR funcionando*

APROVADO

Ping inter-sub-rede — PC1 -> PC3 (192.168.100.94) — 3/4 — OK

`C:\> ping 192.168.100.94`

✔ *Reply from 192.168.100.94 — TTL=126 — atravessa MAR e VEL*

APROVADO

Ping inter-sub-rede — PC1 -> PC4 (192.168.100.126) — 3/4 — OK

`C:\> ping 192.168.100.126`

✔ *Reply from 192.168.100.126 — TTL=126 — conectividade total confirmada*

APROVADO

```
C:\>ping 192.168.100.62

Pinging 192.168.100.62 with 32 bytes of data:

Request timed out.
Reply from 192.168.100.62: bytes=32 time<1ms TTL=127
Reply from 192.168.100.62: bytes=32 time<1ms TTL=127
Reply from 192.168.100.62: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.100.62:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.100.94

Pinging 192.168.100.94 with 32 bytes of data:

Request timed out.
Reply from 192.168.100.94: bytes=32 time=1ms TTL=126
Reply from 192.168.100.94: bytes=32 time=1ms TTL=126
Reply from 192.168.100.94: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.100.94:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 192.168.100.126

Pinging 192.168.100.126 with 32 bytes of data:

Request timed out.
Reply from 192.168.100.126: bytes=32 time=1ms TTL=126
Reply from 192.168.100.126: bytes=32 time=11ms TTL=126
Reply from 192.168.100.126: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.100.126:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 11ms, Average = 4ms
```

Pings de PC1 para PC2 (.62), PC3 (.94) e PC4 (.126) — 3/4 pacotes (1 ARP timeout = normal)

■ O primeiro pacote retorna Request timed out — normal no Packet Tracer (processo ARP). Os 3 pacotes seguintes respondem com sucesso (25% loss esperado).

8. Traceroute — PC1 ate PC4

Hop 1 — MAR (192.168.100.1) — gateway da sub-rede de PC1
`C:\> tracert 192.168.100.126`
✔ *0 ms — primeiro salto correto pelo gateway MAR Gi0/0*
APROVADO

Hop 2 — VEL Serial (192.168.100.226) — link WAN /30
`Traversal do link serial ponto-a-ponto`
✔ *0-1 ms — atravessa o link WAN MAR -> VEL via 192.168.100.224/30*
APROVADO

Hop 3 — PC4 (192.168.100.126) — destino final alcancado
`Sub-rede 192.168.100.96/27`
✔ *1 ms — 3 hops totais confirmam o caminho de roteamento esperado*
APROVADO

```
C:\>tracert 192.168.100.126

Tracing route to 192.168.100.126 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.100.1
  2   0 ms      1 ms      0 ms      192.168.100.226
  3   1 ms      0 ms      0 ms      192.168.100.126
```

tracert de PC1 (192.168.100.30) ate PC4 (192.168.100.126) — 3 hops: MAR > VEL > PC4

✔ Traceroute confirma o caminho correto: PC1 -> MAR (Hop1) -> VEL/WAN (Hop2) -> PC4 (Hop3)

■. Resumo dos Resultados

Teste	Resultado	Status
Calculo sub-redes /27	30 hosts utilizaveis | 8 sub-redes possiveis	APROVADO
VLSM — Link WAN /30	2 hosts utilizaveis — ideal para serial ponto-a-ponto	APROVADO
Configuracao IPs nos PCs	4 PCs com IP estatico, mascara e gateway corretos	APROVADO
Running-config MAR	Gi0/0 .1 | Gi1/0 .33 | Se2/0 .225 | rotas estaticas	APROVADO
Running-config VEL	Gi0/0 .65 | Gi1/0 .97 | Se2/0 .226 | rotas estaticas	APROVADO
Rotas estaticas MAR	S: .64 e .96 via 192.168.100.226	APROVADO
Rotas estaticas VEL	S: .0 e .32 via 192.168.100.225	APROVADO
IP Interface Brief MAR	Gi0/0, Gi1/0, Se2/0 — up/up	APROVADO
IP Interface Brief VEL	Gi0/0, Gi1/0, Se2/0 — up/up	APROVADO
Ping PC1 -> PC2 (.62)	3/4 pacotes — 1 ARP timeout (normal)	APROVADO
Ping PC1 -> PC3 (.94)	3/4 pacotes — atravessa MAR e VEL	APROVADO
Ping PC1 -> PC4 (.126)	3/4 pacotes — conectividade total	APROVADO

Teste	Resultado	Status
Traceroute PC1 -> PC4	3 hops: MAR > VEL > PC4 — caminho correto	APROVADO

Conclusao: Todos os 13 testes foram aprovados. O laboratorio demonstra com sucesso o dominio de sub-redes IPv4 no nivel CCNA — divisao de rede /24 em sub-redes /27 (VLSM), uso de /30 para link WAN, configuracao de roteamento estatico bidirecional entre dois roteadores, verificacao de interfaces e conectividade completa entre todos os hosts das quatro sub-redes.